

Tabular Data

- Data in the form of a **table** consisting of a rectangular collection of **rows** and **columns** has become the most important way that data is structured, organized, and used today.
- In a data table, each **row** represents a **record**, and each **column** represents a **field or attribute**:

Last Name	First Name	Class Level	Home State	Major	GPA
Bania	Kenny	1	New York	Communications	1.20
Benes	Elaine	3	New York	Business	3.30
Sideshow	Bob	4	California	Drama	4.00
Bouvier	Patty	2	California	Speech	2.22

← record
← record
← record
← record

↑ ↑ ↑ ↑ ↑ ↑
fields (attributes)

Tabular Data

- **Spreadsheets** provide an outstanding way
 - to **view** the same tabular data that we may be using in other software

	A	B	C	D	E	F	
1	Last Name	First Name	Class Level	Home State	Major	GPA	
2	Bania	Kenny	1	New York	Communications	1.20	← record
3	Benes	Elaine	3	New York	Business	3.30	← record
4	Sideshow	Bob	4	California	Drama	4.00	← record
5	Bouvier	Patty	2	California	Speech	2.22	← record

↑ ↑ ↑ ↑ ↑ ↑

fields (attributes)

- to **perform operations** on the same tabular data that we may be using in other software, thereby
 - enabling us to **preview** those operations
 - and providing one way to **check the results** of those operations in other software.



Solidifying Our Spreadsheet Abilities

- For this reason, in this course, it will be enormously valuable for us to solidify our **spreadsheet abilities!**
- We will build our way up from the important **foundations**, then work our way up to more **advanced techniques** and operations that involve **functions**, **visualization**, and more specifically **database-oriented** operations.
- **Trust me**: starting by solidifying the **foundational** concepts and operations will be crucial for this.
- I teach this material in a somewhat distinct way, and many of my previous **LIS 440** students found this to be some of their favorite and most **valuable** material they learned in this course!

- A software **spreadsheet** contains lettered **columns**

	A	B	C	D	E	F
--	---	---	---	---	---	---

and numbered **rows**

	1	2	3	4	5
--	---	---	---	---	---

- These columns and rows intersect to form "cells."
- Each cells is addressed using its **column & row**.

example: cell **B3**

	A	B
1		
2		
3		
4		

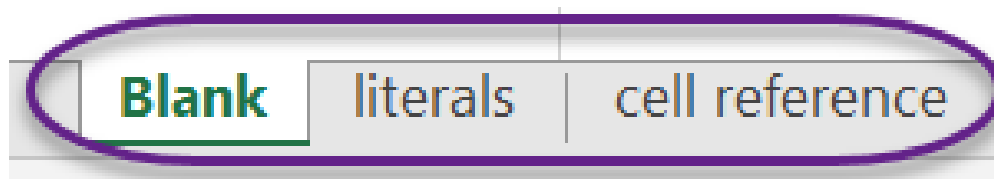
Excel "Worksheets" and Workbooks"



- In Excel, a **spreadsheet** is known as a "worksheet."
 - For example, the current worksheet is named "**Blank**":



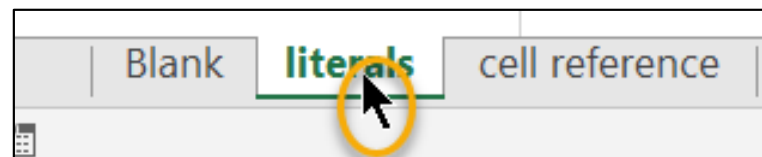
- A workbook is comprised of multiple worksheets."



- Each Excel file contains one "workbook."
 - Thus, each Excel **file** can contain **multiple worksheets**.

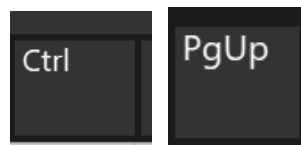
To **navigate** between worksheets:

- click your mouse pointer on the name **tabs** at the bottom of worksheets



- Or use these **keyboard shortcuts**:

Ctrl-Page Up



Ctrl-Page Down



- If some of the tabs are not visible on the screen
 - Click the **ellipses** or the **arrow** buttons next to the tabs:



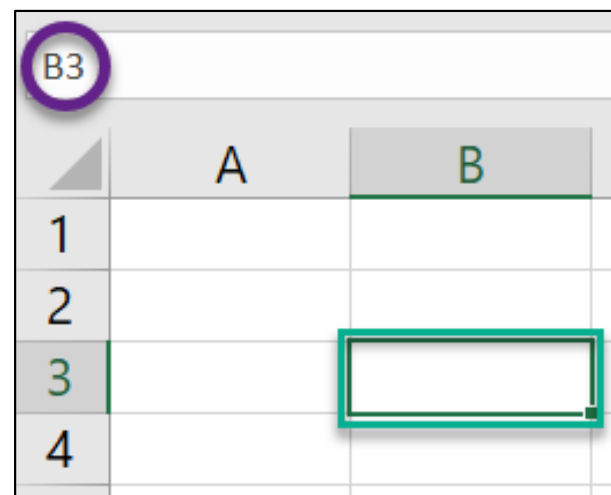
- Select cells using the **mouse pointer** or **arrow keys**.

Keyboard Shortcuts:

- **Page Up/Page Down** – up/down one **screen**
- **Home** – jumps to column **A** in the current row
- **Ctrl-Home** – jumps to cell **A1**
- **Ctrl-End** – jumps to lower-rightmost cell with data in it
- **Ctrl-G** or **F5** – enter cell address to select

Selecting a cell makes it the "**active cell**."

Here, **B3** is the active cell:



The image shows a portion of an Excel spreadsheet. The column headers are A and B, and the row headers are 1, 2, 3, and 4. Cell B3 is highlighted with a thick green border, indicating it is the active cell. A purple circle with the text 'B3' is positioned above the active cell. The cell itself is empty.

	A	B
1		
2		
3		
4		

- After typing content into a cell, you can press:

Enter

- This enters the content into the cell and then the cell beneath it becomes the active cell.

Ctrl-Enter

- This will enter the content and keep the current cell the active cell.

Tab

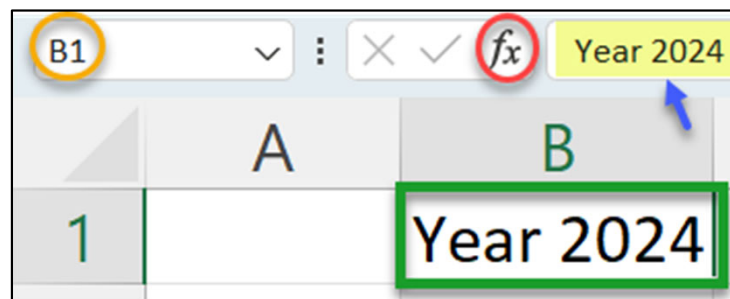
- This will enter the content and make the cell to the **right** the active cell.

An arrow key

- This will enter the data and then make the cell in the direction corresponding to the arrow the active cell.

To **change** the data in a cell, do **any** of the following:

- **Select** the cell and **type in** the desired content.
- **Double-click** the cell and edit the current cell content.
- Press **F2** and then edit the current cell content.
- Select the cell and click on the **formula bar**

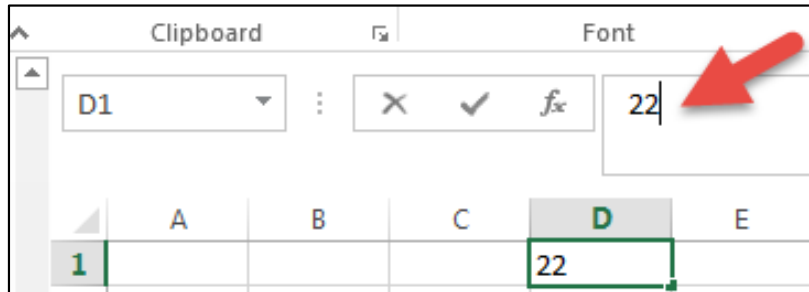


and then edit the current cell contents there.

- After editing, press **Enter**, **Ctrl-Enter**, or **Tab**.
- To **exit edit mode without** saving edits, press the **Esc** key.

The "Double View" of Cells

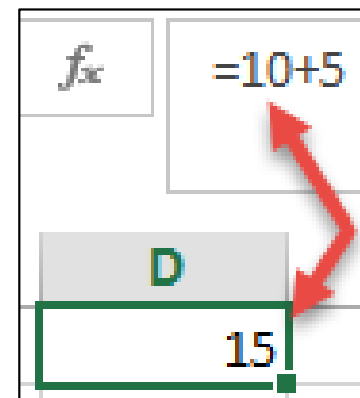
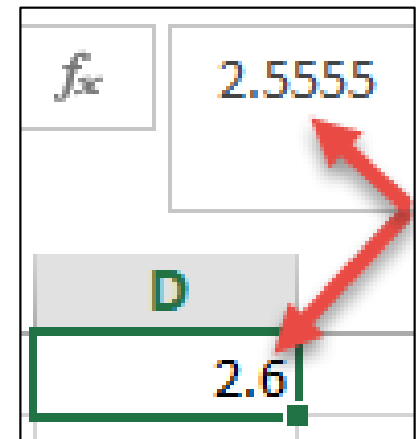
- the **formula bar** displays the actual content of a cell.



- In the cell itself, what is **displayed** is the **result** of
 - formatting** choices applied to the cell
 - e.g. number of **decimal places**

and/or

- a **formula** that is the actual content of the cell.



“Literal” Values

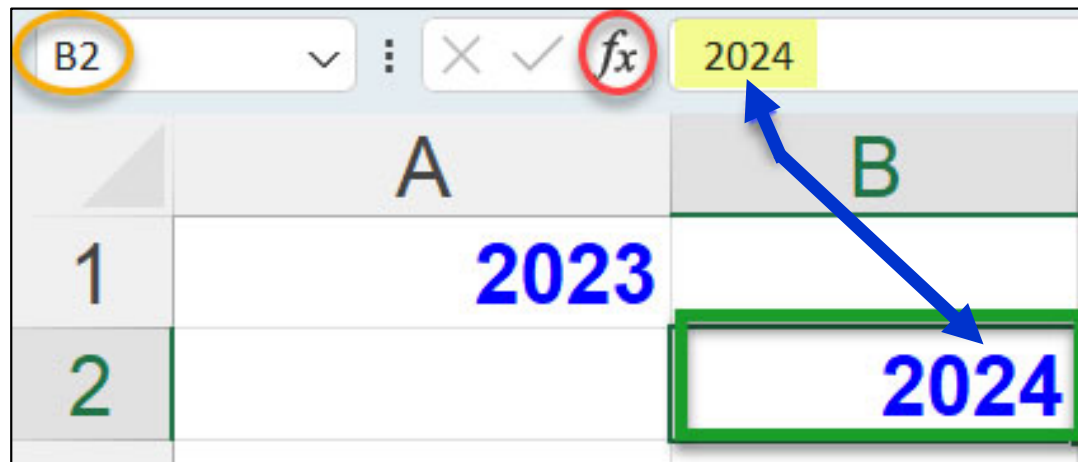
literals

- “**Literal**” values are cell entries that do not **change** unless someone **edits** them.

numeric literal: **2024**

text literal: **Jeff**

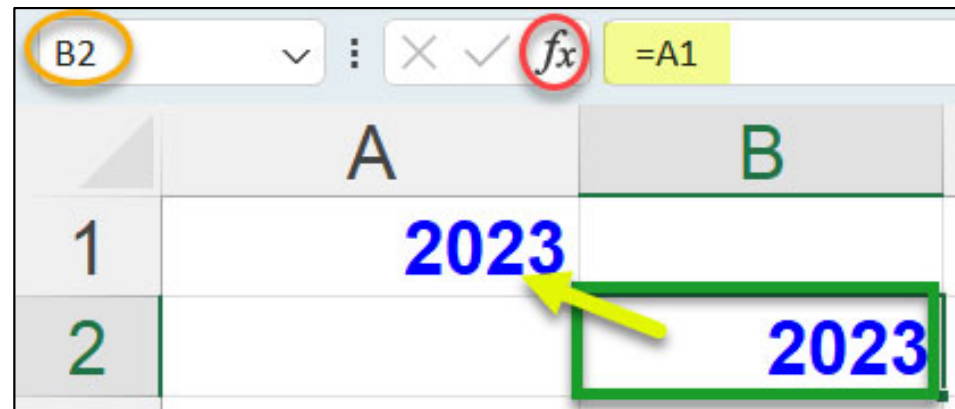
- If we click to select a cell containing a “**literal**” value, then “formula bar” typically displays the same value as that **displayed** in the **cell** itself:



Cell References

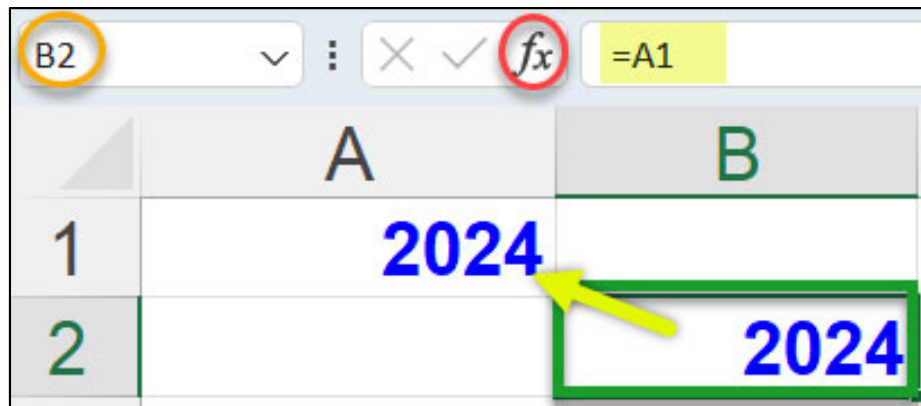
cell reference

- A **cell** can be set to refer to the value in another cell.

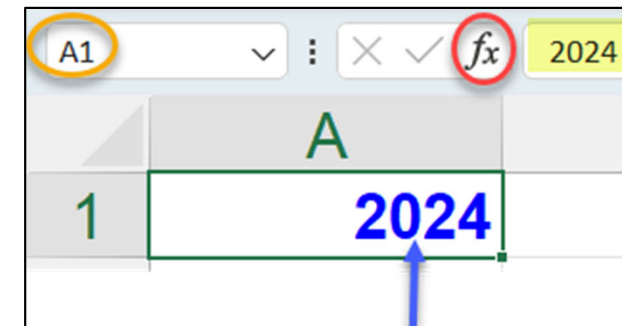


	A	B
1	2023	
2		2023

- Here, if the value of cell **A1** changes, then the value of cell **B2** is updated automatically:



	A	B
1	2024	
2		2024



	A	B
1	2024	
2		2024

- This makes the cell **B2** a "dependent" of cell **A1**.
- This makes the cell **A1** a "precedent" of cell **B2**.

Formulas in Cells

simple formula



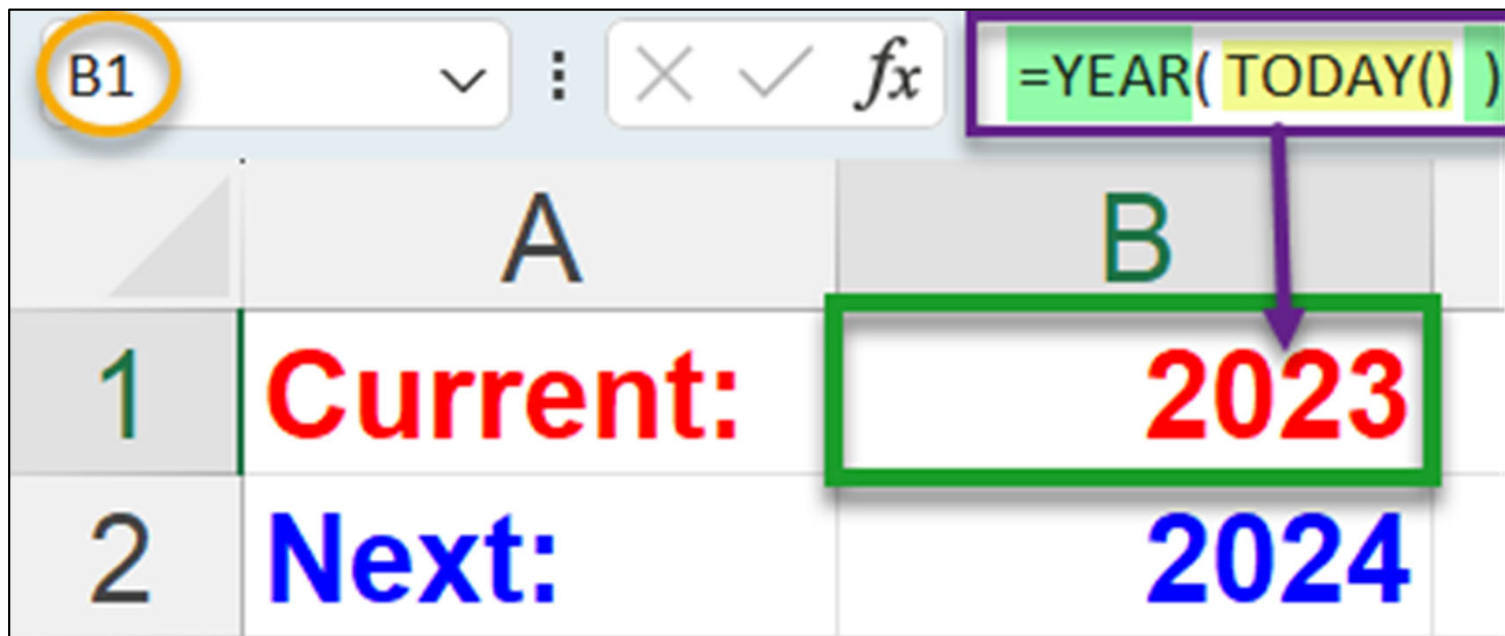
- Cells can contain **formulas**.
- Excel **formulas** must begin with an **equals sign**: **=**
- When we click to select a cell containing a **formula**, the **formula bar** displays that formula, while the cell itself displays the **result** of that formula.

B2	⌵	⋮	✕	✓	<i>fx</i>	=B1+1
		A				B
1		Current:				2023
2		Next:				2024

Formula and Function in Cells

formula with function

- Cells can also contain **calls to functions**.
- Excel has a large "library" of functions.
- These functions return useful values related to the kind of function it is.
- Here, the **TODAY()** function returns the current date, and the **YEAR()** function pulls the year out of it, with the result placed in cell **B1**:



	A	B
1	Current:	2023
2	Next:	2024

Avoid Literals in Formulas

Tip Calculator



- Avoid using a **literal** value in a formula!

	A	B	C
1	Bill:	20.44	
2			
3	Tip:	5.11	

- Literals in formulas are unexplained, “**magic numbers.**”
 - For example, why are we multiplying the bill by **0.25** ?
Apparently, it is because of a **25%** tip, but this isn't at all clear...
- Literals “**hard coded**” into formulas might need to **change**!
 - This forces you to **edit the formula** every time you need to change the assumption.
 - For example, what if we want tip an amount other than **25%**?

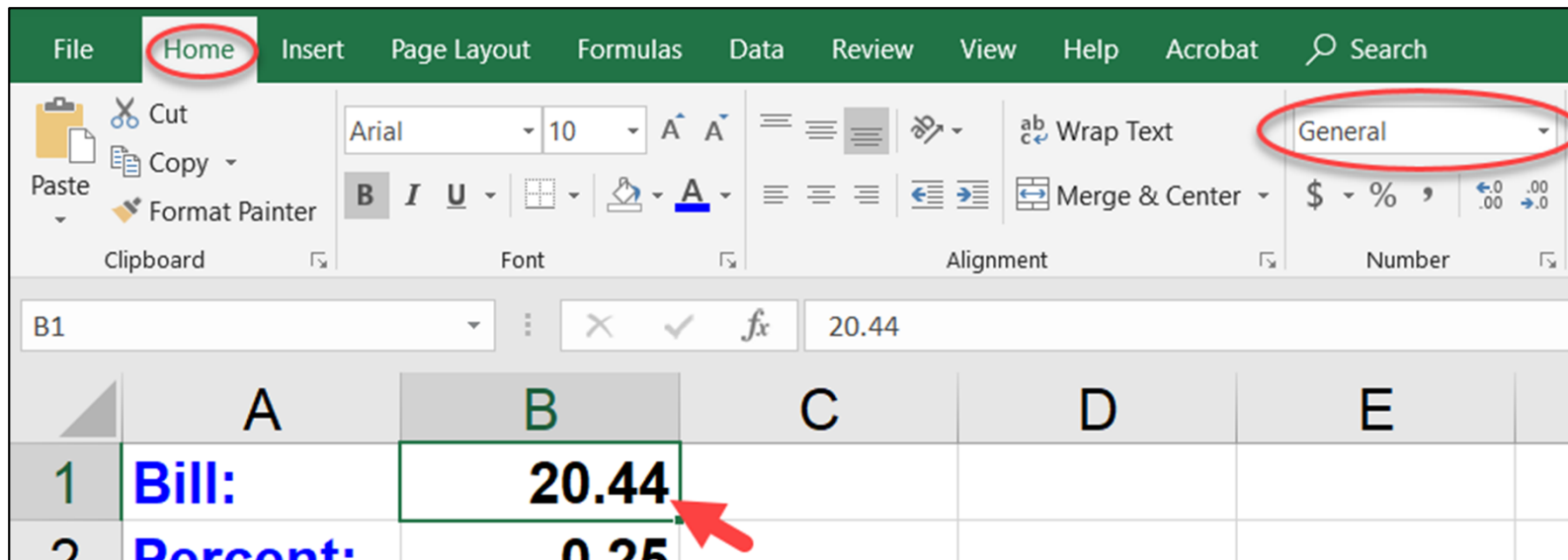


- Use a **cell address** instead of a **literal** in a formula!
 - This allows us to **isolate** this assumption
 - For example, the **tip percentage**.

- This allows to **experiment** with this assumption.
 - This is sometimes known as "**What-If Analysis**."
 - We can try **22%** or **18%**, etc., without having to edit the formula.
- We can also now **label** this assumption:

- Clearly identified this way, it is no longer a "magic number."

- The way a value is **displayed** in a cell depends upon the **format** settings for that cell.
- The current format for cell **B1** is "**General**"
 - This is the default format, indicating no special formatting.

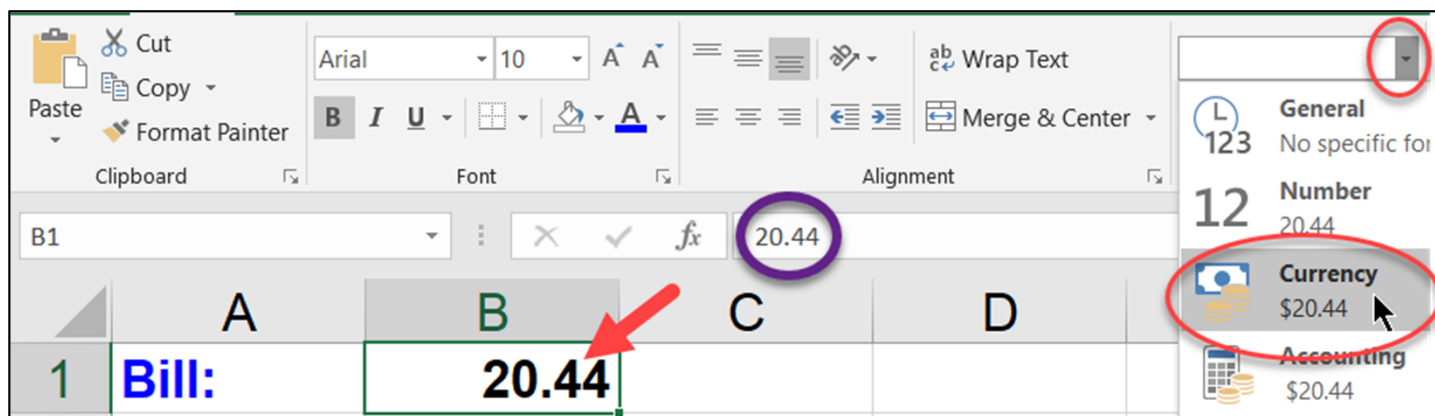


Formatting Monetary Amounts

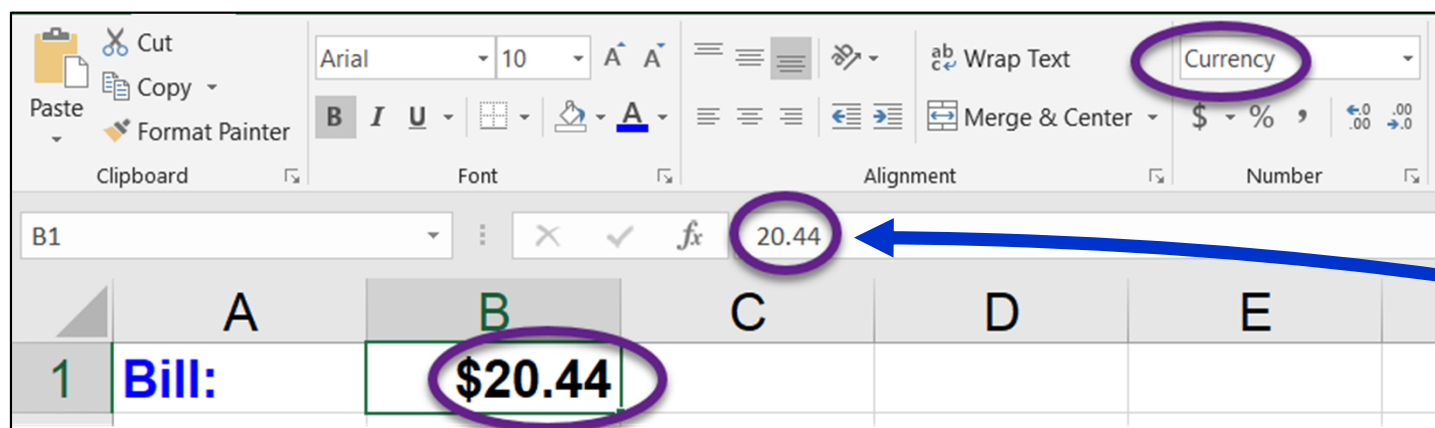
Tip Calculator



- We can change the format for cell **B1** to **Currency**:



- In Currency format, the numeric value in this cell, **20.44**, now looks more like a dollar amount:



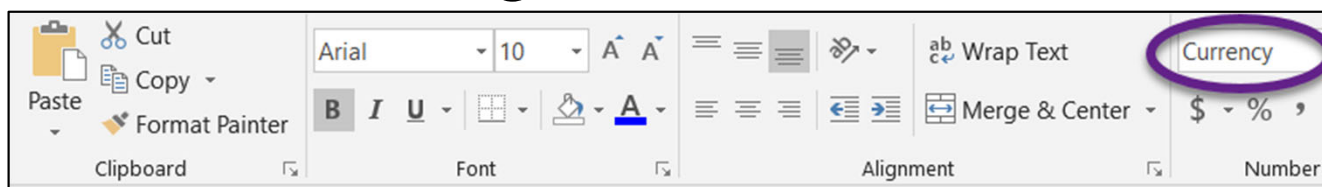
However, if we look at the formula bar, we see that the actual number value in the cell has not changed.

Formatting Monetary Amounts

Tip Calculator

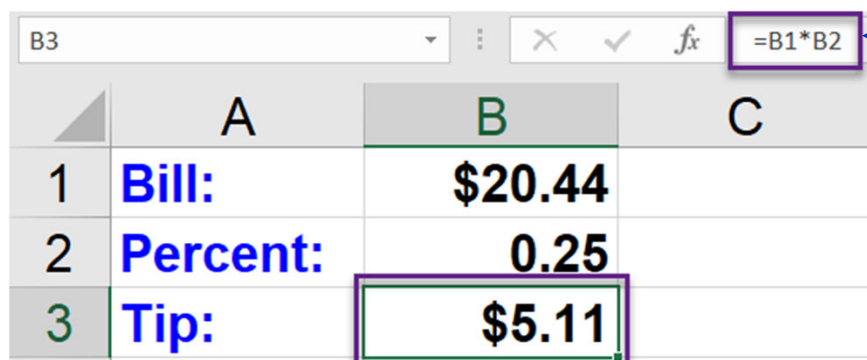


- If we also change the format for cell **B3** to **Currency**...



	A	B	C	D	E
1	Bill:	\$20.44			
2	Percent:	0.25			
3	Tip:	\$5.11			

then the cell now looks more like a monetary value..

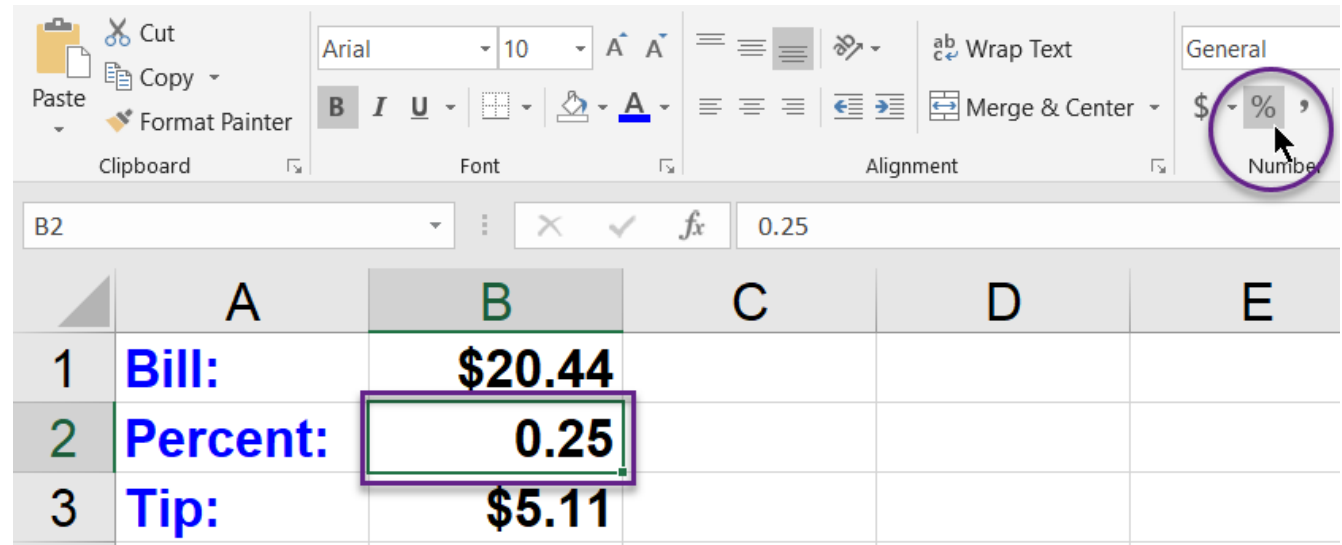


	A	B	C
1	Bill:	\$20.44	
2	Percent:	0.25	
3	Tip:	\$5.11	

However, if we look at the formula bar, we see that the formula in this cell has not changed.

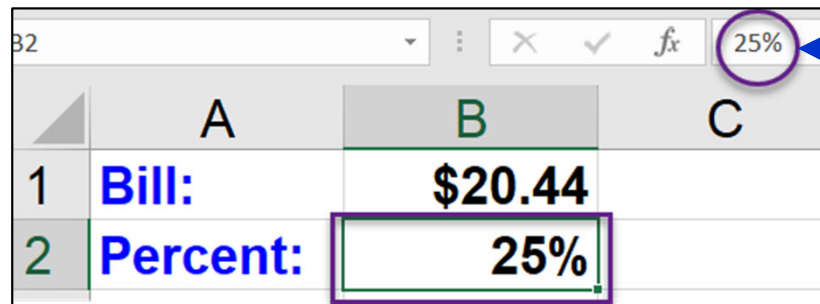
Formatting Percentages

- We can change the formatting for cell **B2** to **Percentage**:



	A	B	C	D	E
1	Bill:	\$20.44			
2	Percent:	0.25			
3	Tip:	\$5.11			

- In **Percentage** format, the numeric value in this cell now more like a percentage...

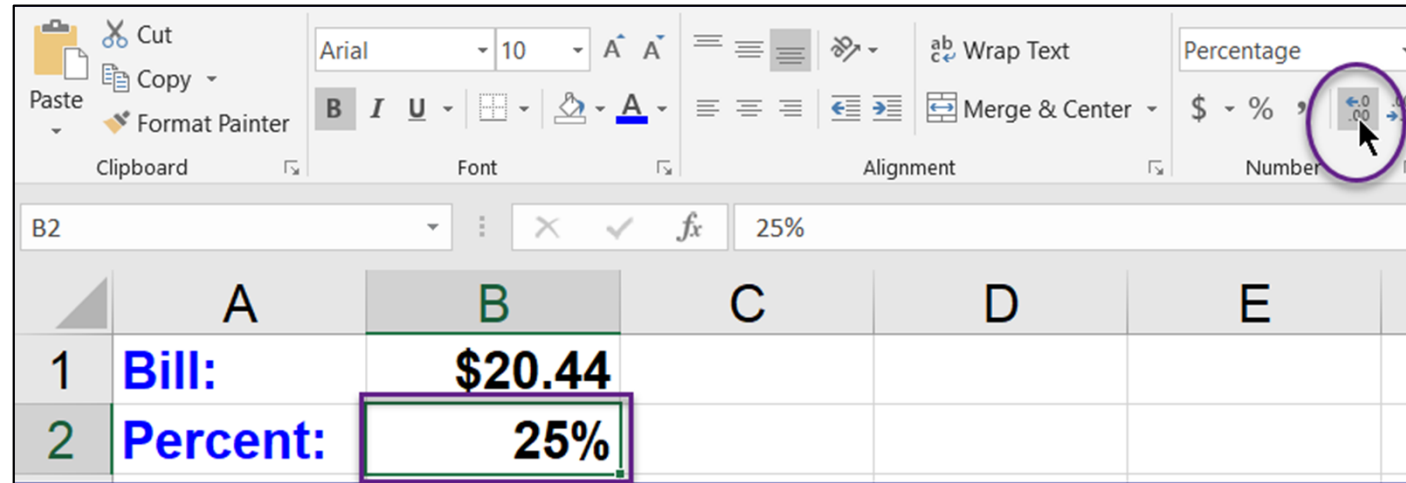


	A	B	C
1	Bill:	\$20.44	
2	Percent:	25%	

If we look at the formula bar, we see that Excel uses the **%** symbol in the formula, too. Excel also allows us to enter percentages using this symbol.

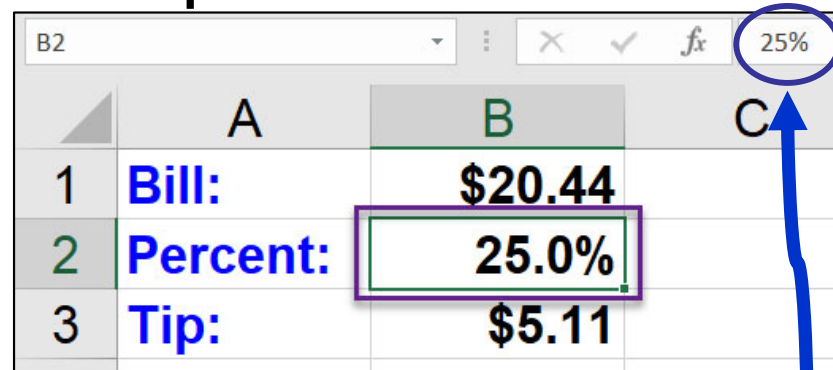
Formatting Decimals

- We can also **increase** the number of **decimal places** for cell **B2**:



	A	B	C	D	E
1	Bill:	\$20.44			
2	Percent:	25%			

- The numeric value in this cell is now displayed with an additional decimal place...

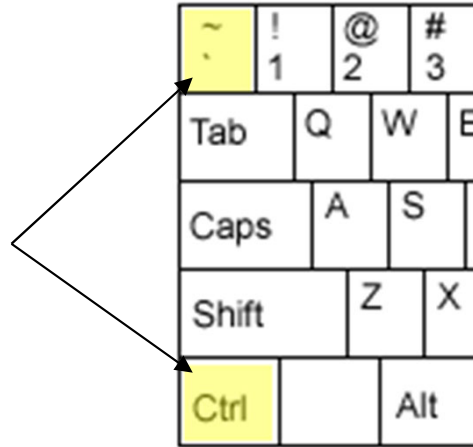


	A	B	C
1	Bill:	\$20.44	
2	Percent:	25.0%	
3	Tip:	\$5.11	

However, if we look at the formula bar, we see that the percentage value that is actually in the cell has not changed.

Displaying Actual Cell Contents & Formulas

- **Ctrl** and **~** pressed simultaneously...



toggles between displaying **formulas/formatting** and displayed **values**.

- Or, on the **Formulas** tab, click **Show Formulas** :

